

Minimal-Shot AV Evidence Deck

Architecture, SOTA Submission Results, and CoRL Transfer Diagnostics

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Waymo Open Dataset E2E Challenge / SOTA + CoRL 2027 branch

How to Read This Deck

This is not only a CoRL paper presentation.

It has two audiences:

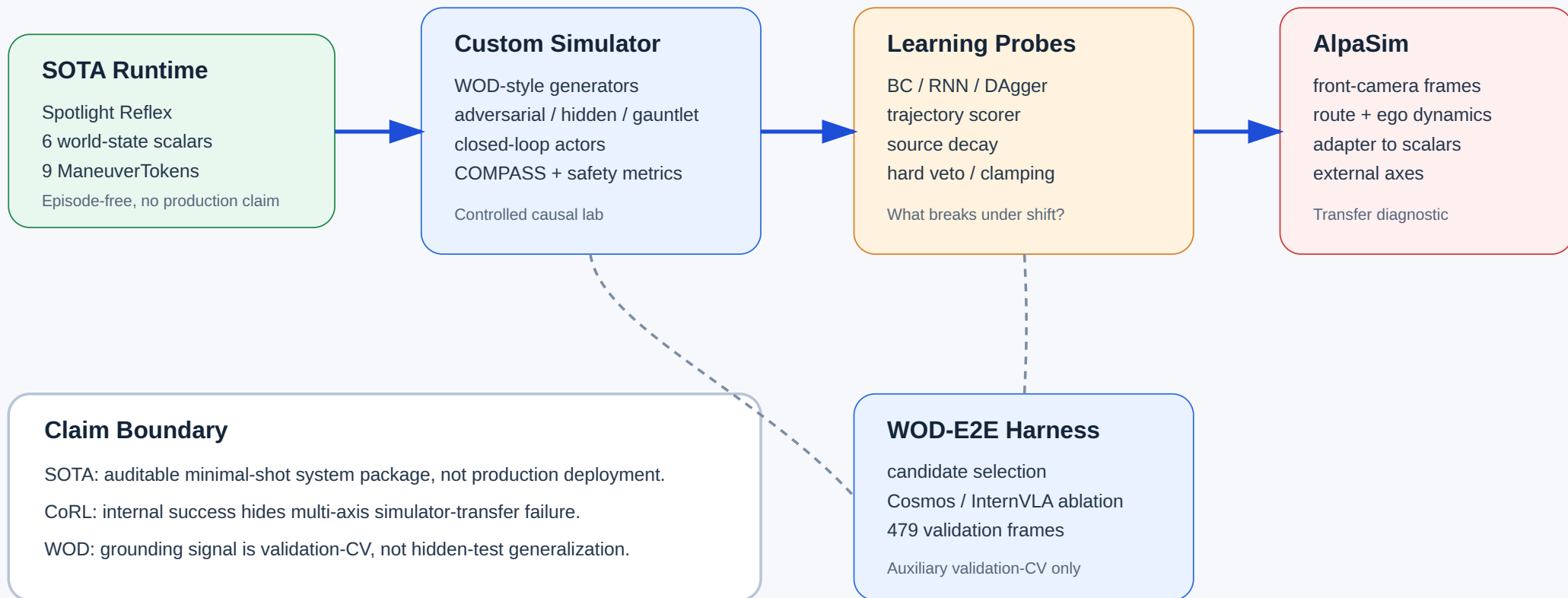
Audience	What they need	Claim boundary
SOTA auditors	architecture, reproducibility, packaged evidence	submission candidate, not production
CoRL reviewers	mechanistic transfer diagnosis	diagnostic paper, not method dominance

The architecture is the bridge between both.

The Whole System in One Graph

Minimal-Shot AV: One Repo, Two Claims

SOTA submission is the packaged system. CoRL paper is the transfer diagnostic built on the same infrastructure.



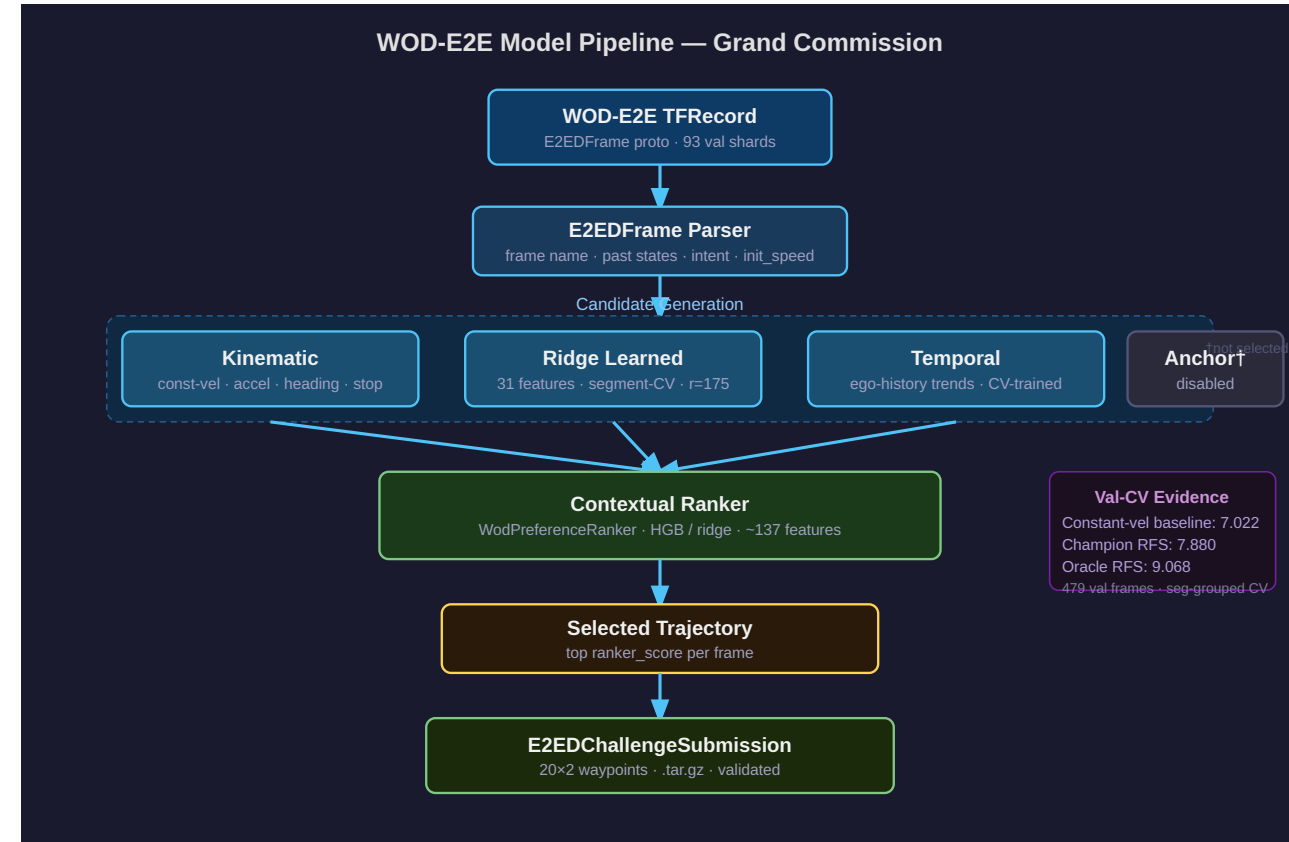
What We Actually Built

Component	Role	Why it matters
Spotlight Reflex runtime	geometry-driven token selector	auditable minimal-shot policy
Custom simulator	closed-loop long-tail stress lab	controlled causal experiments
Dagger / scorer probes	learned baselines	tests what imitation learns
AlpaSim adapter	external transfer interface	exposes proxy-state mismatch
WOD-E2E harness	real-data selector benchmark	tests frozen grounding priors

Runtime Architecture: Spotlight Reflex

Runtime path:

- obstacle and route geometry become scalar world-state
- 9 ManeuverTokens are generated
- reference rules score token futures
- selected token is executed by the controller



Token Interface

The same token interface is used across the SOTA runtime, learned policies, and AlpaSim variants.

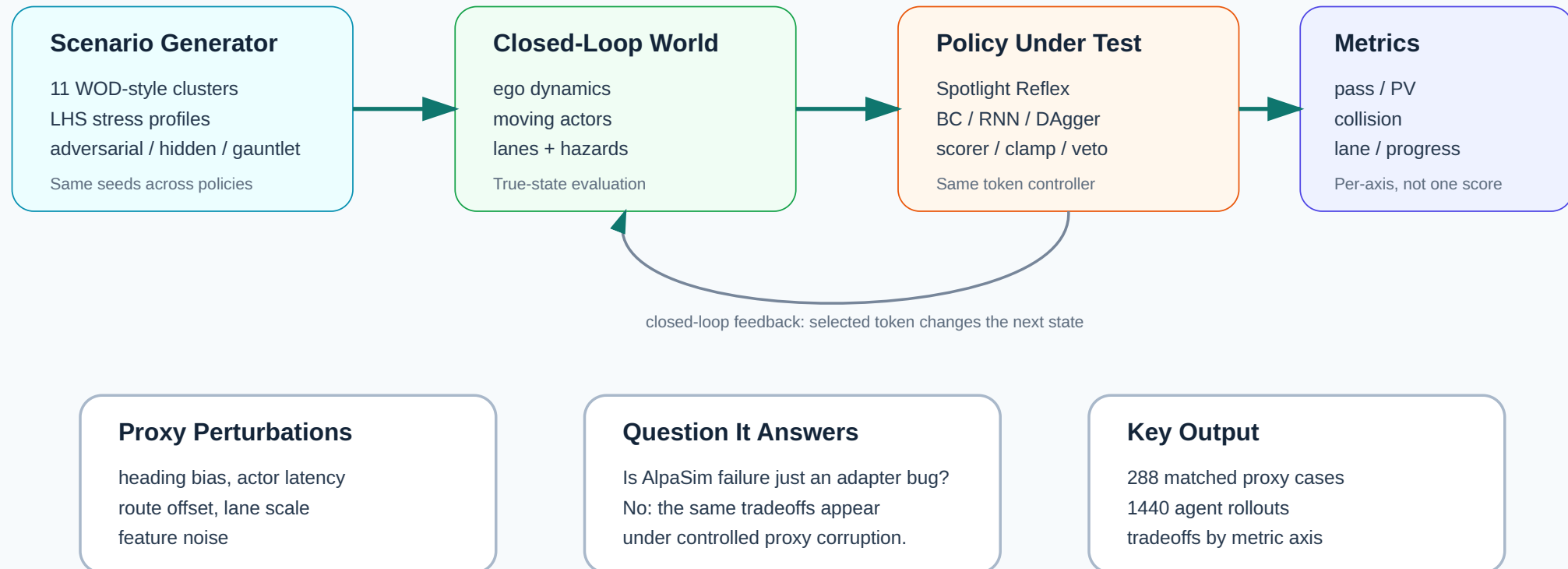
Token family	Examples	Function
stop / crawl / slow-yield	stop, crawl, slow_yield	longitudinal safety
nominal progress	maintain	forward route completion
mild lateral evasion	nudge_left, nudge_right	local clearance
strong lateral evasion	evasive_left, evasive_right	emergency separation
recovery	lane_recover	return to lane center

This shared interface is why we can isolate selection failure from candidate-set failure.

Simulator Architecture

Custom Simulator: Why It Exists

It is a controlled stress lab where state, proxy corruption, policy, and metrics are independently manipulable.

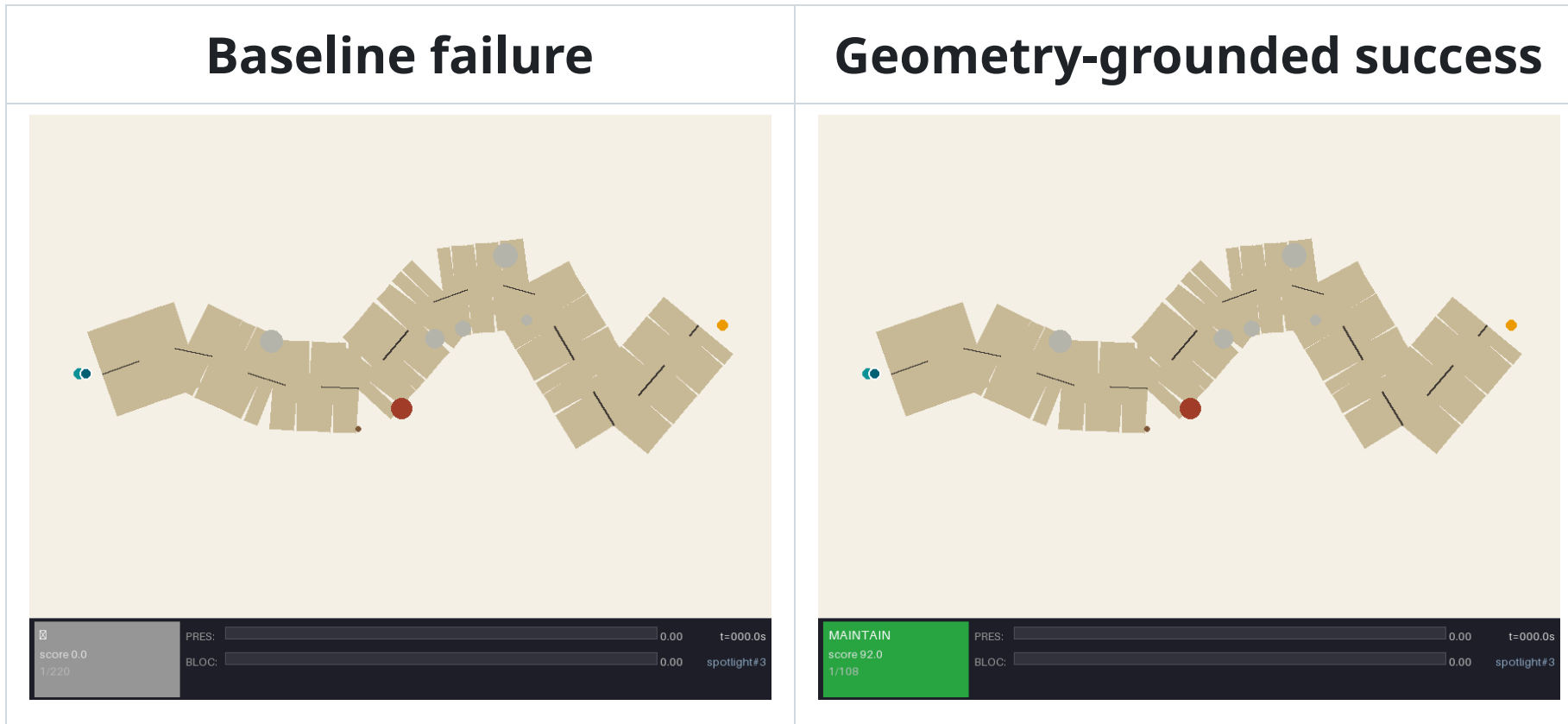


What the Simulator Lets Us Test

Variable	Controlled how
scenario difficulty	WOD-style clusters, gauntlet, adversarial, hidden
distribution shift	Latin-hypercube sampled generator profiles
policy class	Spotlight, BC, RNN, DAgger, scorer, veto
proxy mismatch	heading bias, actor latency, route offset, lane scale, noise
outcome axis	collision, PV, lane, offroad, progress, distance-to-GT

This is why the simulator results are useful for CoRL: they isolate mechanism before external validation.

Visual Intuition



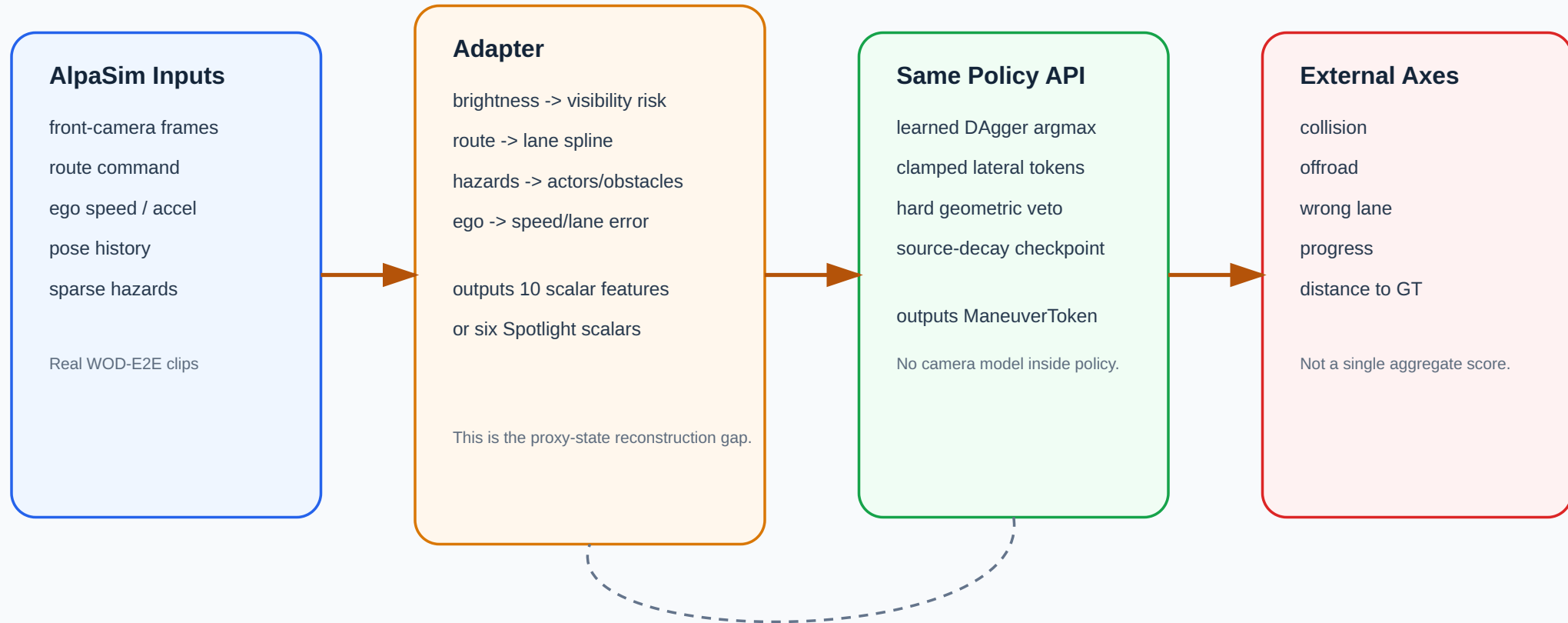
Same scenario family: wrong-way actor under low visibility.

The baseline extrapolates into the hazard. Spotlight chooses an evasive token

AlpaSim Transfer Architecture

AlpaSim Transfer Stack

The policy is held fixed. Transfer happens through the observation adapter and execution interface.



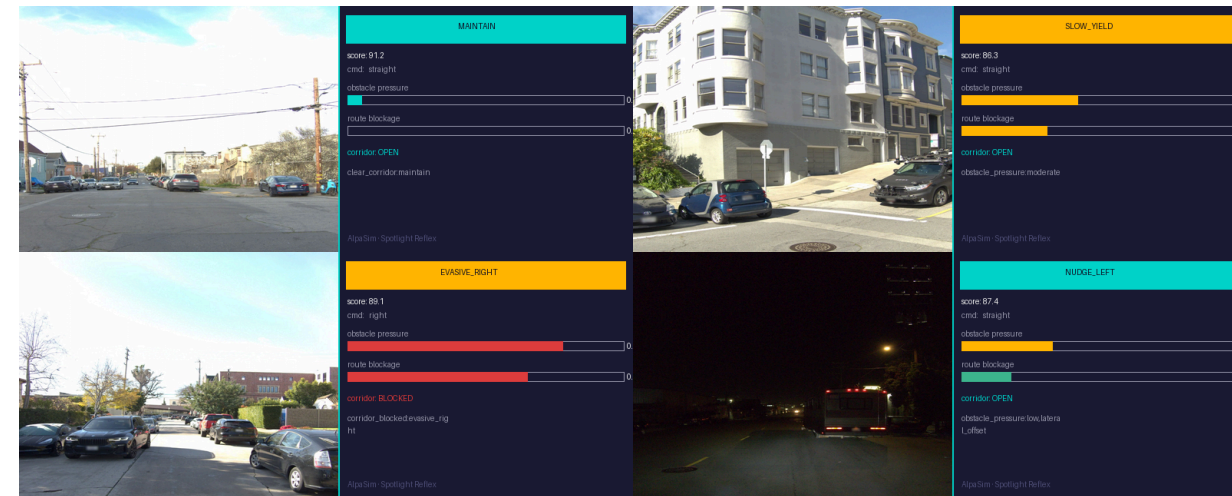
Selection failures feed back through execution, not through offline accuracy.

AlpaSim Adapter View

What this slide is showing:

- real WOD-E2E front-camera frames
- adapter reconstruction of route and hazard signals
- same downstream scalar policy API

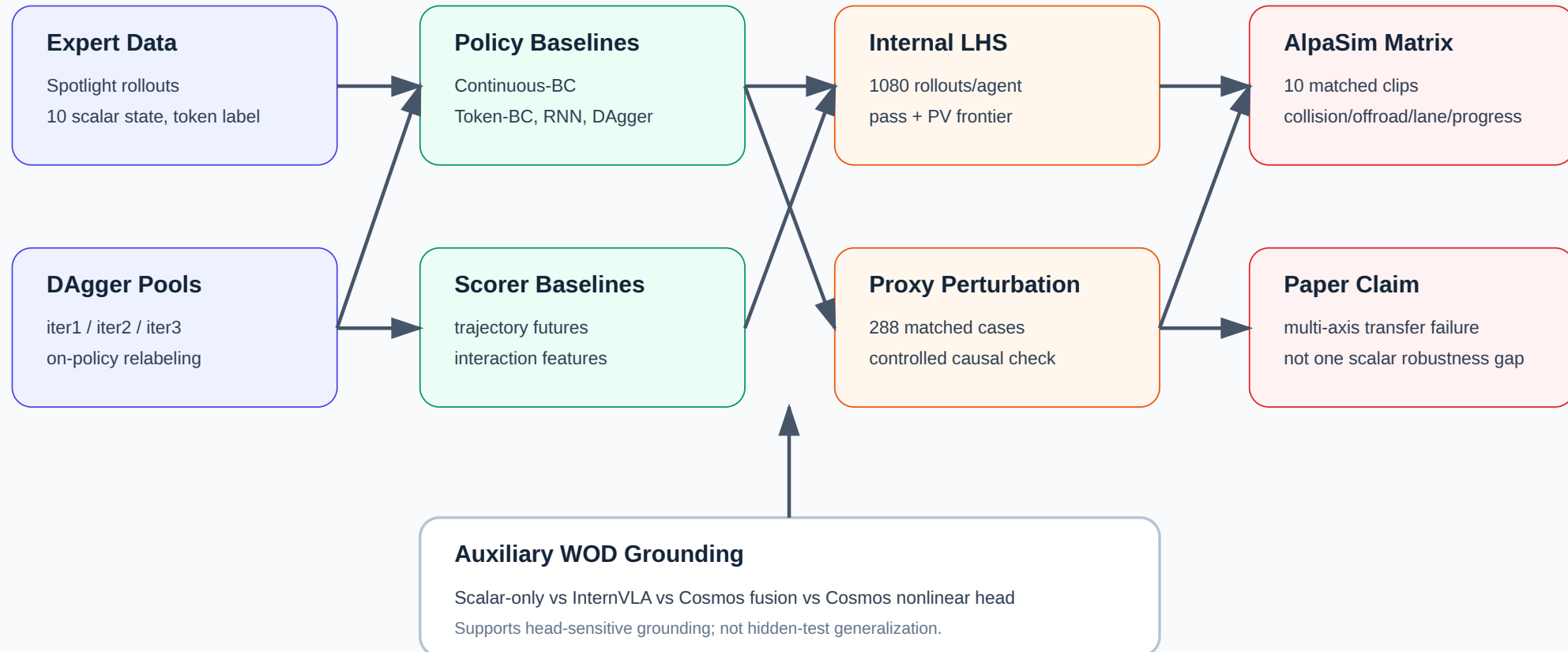
This is the proxy-state reconstruction gap made visible.



Experiment Dependency Graph

Experiment Dependency Graph

The presentation should make clear which results depend on which artifacts.



SOTA Submission Status

Gate	Status
Grand archive ready	pass
Minor archive ready	pass
Minimal-shot integrity	pass
Judging-criteria evidence	pass
Production no-go boundary declared	pass
WOD symbolic-trust risk-control boundary	pass

No blockers are reported in `artifacts/final_submission_readiness_audit.json`.

Boundary: `submission_candidate_not_hidden_test_not_production`.

SOTA Primary Result

Closed-loop simulation headline from the SOTA evidence package:

Suite	Runs	Collisions	Pass
WOD-style, all 11 clusters	110	0	100%
Compositional OOD	60	0	100%
Adversarial	60	0	100%
Hidden holdout	60	0	100%
Gauntlet	60	0	60%
Total	350	0	93.1%

COMPASS: **9.137 / 10** over 700 ranked runs.

Source: [README.md](#) | [artifacts/compass_evidence_report.json](#)

SOTA Baseline Comparison

Matched gauntlet comparison: same 420 scenarios, same seeds.

Policy	Pass rate	Collision rate
Baseline, no world-state reasoning	2.1%	20.5%
Spotlight Reflex	57.6%	7.9%

Interpretation:

- the SOTA contribution is an auditable runtime path
- the policy is not selected by scenario ID
- the failure boundary is measured, not hidden

WOD-E2E Auxiliary Benchmark

479 WOD-E2E validation frames, segment-grouped CV.

Selector	RFS	Folds	Role
Waymo baseline	7.022	-	reference
Local baseline	7.131	-	reference
Gate-only	7.803	5	selector baseline
RFF direct policy	7.834	5	learned baseline
GPU MLP + Cosmos 64d	7.845	5	champion validation-CV
Oracle selector	9.264	5	upper bound

This is auxiliary evidence, not a hidden-test or zero-shot WOD claim.

CoRL Research Question

Internal success looked strong.

But did it transfer?

The question became:

When a learned token policy succeeds inside the training simulator, which parts of its behavior survive a different observation and execution interface?

That is the CoRL paper: a diagnostic of transfer-axis disagreement.

Internal LHS Result

Held-out Latin-hypercube sweep: 12 profiles, seeds 1-10, Gauntlet / Adversarial / Hidden.

Each agent: 1080 closed-loop rollouts.

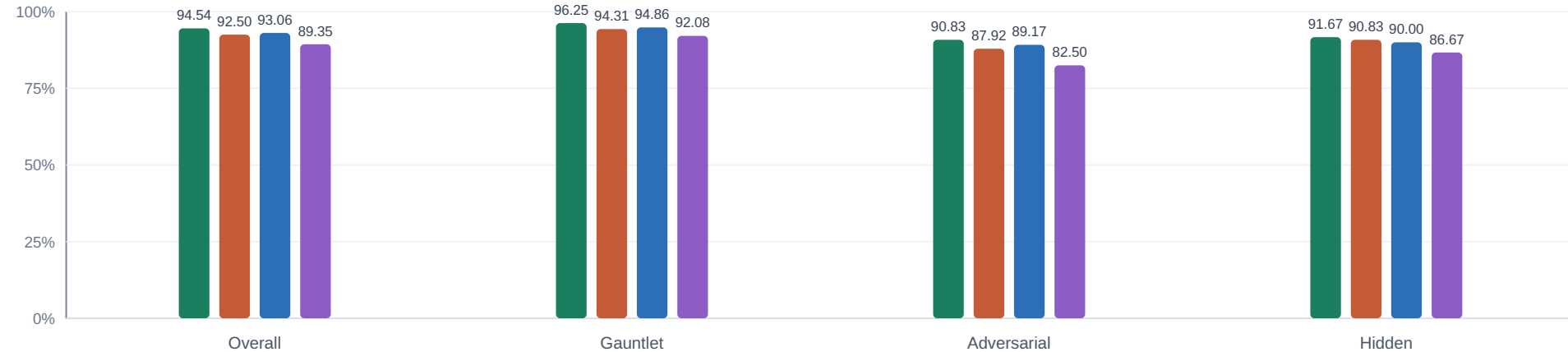
Agent	Overall pass / PV	Gauntlet	Adversarial	Hidden
Continuous-BC	77.69 / 3.52	77.50 / 4.31	75.42 / 2.08	83.33 / 1.67
Token-BC	76.85 / 4.26	77.64 / 4.86	71.67 / 2.92	82.50 / 3.33
Token-RNN-BC	78.43 / 2.96	79.17 / 2.92	74.58 / 2.50	81.67 / 4.17

Dagger Aggregation Is Non-Monotonic

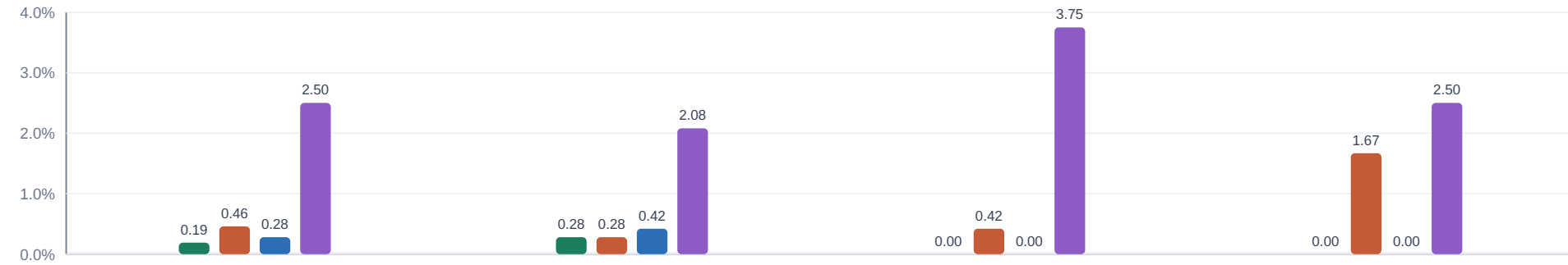
Dagger Aggregation Ablations on Held-Out Latin-Hypercube Sweep

Two-step Dagger remains the frontier. Source decay partially recovers the three-step collapse; removing inverse-frequency weights breaks closed-loop safety.

Pass Rate



Collision Rate



Why Trajectory Scoring Was Not Enough

We tested trajectory-informed scorers over candidate futures with:

- curvature, heading deltas, path length, jerk proxies
- minimum dynamic clearance and time-to-closest-approach
- signed lateral and longitudinal miss distances

Policy	Gauntlet pass / PV	Adversarial	Hidden
Interaction scorer, learner states	86.81 / 1.94	85.00 / 1.67	89.17 / 1.67
Interaction scorer, merged states	88.75 / 1.67	82.92 / 2.50	90.00 / 2.50
			91.67 /

AlpaSim Result: Axes Separate

10 matched WOD-E2E clips completed by all four variants.

Model	Collision	Offroad	Wrong lane	Progress	Dist. m	Dist.- GT
Raw DAgger iter2	0.600	0.900	0.700	0.034	61.98	13.86
Clamped lateral	0.700	0.500	0.200	0.384	59.85	5.21
Hard veto hybrid	0.800	0.200	0.700	0.837	166.06	22.71
Source decay	0.600	0.900	0.400	0.175	62.92	27.16

One intervention repairs one axis and fails or worsens another.

Interpreting the AlpaSim Matrix

Intervention	Repairs	Worsens or fails
clamping	offroad, wrong-lane, distance-to-GT	collision increases
hard veto	progress, offroad	collision increases, wrong-lane unchanged
source decay	wrong-lane, progress	offroad unchanged, distance-to-GT worsens

The hard-veto log is decisive:

- 1987 / 1990 learned DAgger argmax decisions were vetoed
- progress recovered mostly by collapsing toward geometric scoring
- that is not a clean hybrid win

Controlled Proxy-State Test

The internal simulator reproduces the same effect when only the policy-visible proxy state is corrupted.

Perturbation	Raw p/c/l	Clamp	Hybrid	
clean	0.917/0.083/0.312	0.938/0.062/0.167	0.958/0.042/0.167	0.938,
actor latency	0.604/0.396/0.875	0.604/0.396/0.521	0.854/0.146/0.208	0.833,
route offset	0.938/0.062/0.229	0.875/0.125/0.021	0.896/0.104/0.062	0.938,
feature noise	0.917/0.083/0.917	0.938/0.062/0.146	0.917/0.083/0.125	0.896,

p/c/l means pass / collision / lane-violation rate.

Source: `artifacts/internal_proxy_transfer_medium.md` .

The Mechanism

AlpaSim alone could be dismissed as an adapter artifact.

The proxy test makes the mechanism harder to dismiss:

- same simulator dynamics
- same token library
- same controller
- same matched seeds
- only proxy state is corrupted

The multi-axis tradeoff appears internally and externally.

The Paper Claim

Let each candidate token have a metric vector:

```
m(a) = [collision risk, offroad risk, lane risk, progress loss, tracking error]
```

If a proxy-state perturbation changes action ordering for one metric but not another,

a scalar intervention can improve one axis while worsening another.

The empirical contribution is showing that this happens repeatedly in the AV transfer stack.

Grounding Context: LaST-VLA

Recent VLA work argues for physically grounded latent reasoning.

LaST-VLA uses 3D geometric priors and world-model dynamics to improve autonomous-driving VLA planning.

Our result is complementary:

- grounding signal exists
- but grounding must be evaluated per transfer axis
- aggregate planning scores can hide conflicting improvements

Reference: [LaST-VLA, arXiv:2603.01928](#)

WOD Grounding Probe

Ablation	RFS	Oracle	Regret	Oracle match	Grounding signal
Scalar / geometry only	7.695	9.068	1.373	0.403	none
InternVLA only	7.672	9.225	1.554	0.392	InternVLA
Cosmos + InternVLA linear fusion	7.728	9.208	1.480	0.403	Cosmos + InternVLA
Cosmos 64d nonlinear head	7.845	9.264	1.419	0.390	Cosmos + nonlinear head

Grounding helps only with the right head.

This supports the paper framing, but it is not a hidden test claim

Evidence Strength Audit

Current CoRL audit conclusion: `strong_diagnostic` .

Gate	Current	Target	Status
Internal proxy diagnostic	288 cases	≥ 288 cases	pass
External transfer diagnostic	10 matched scenes	≥ 10 scenes	pass
Larger external scale	10 scenes	≥ 30 scenes	not yet
Uniform positive method	0 external dominance cases	≥ 1	not yet
Grounding prior signal	+0 150 BES	≥ 0	pass

Claim Boundary

Question	Answer
Is the SOTA submission packaged?	yes, readiness audit has no blockers
Is this a production AV safety claim?	no, production audit is no_go
Is WOD 7.845 a hidden-test claim?	no, validation-CV only
Is the CoRL method a uniform winner?	no, external dominance count is 0
Is the CoRL diagnosis supported?	yes, internal proxy + AlpaSim both show tradeoffs

This protects both submissions: SOTA stays auditable, CoRL stays scientifically defensible

Reproducibility Path

SOTA submission audits:

```
artifacts/final_submission_readiness_audit.json  
artifacts/sota_judging_criteria_audit.json  
artifacts/minimal_shot_claim_audit.json  
artifacts/sota_submission_bundles/
```

CoRL paper and audit:

```
docs/corl2027/paper.tex  
docs/corl2027/AUDIT.md  
./scripts/run_corl2027_audit.sh
```

GitHub branch path:

```
https://github.com/amtellezfernandez/minimal-shot-av/tree/CoRL-2027/docs/corl2027
```

Presentation Framing

For SOTA:

A reproducible minimal-shot AV system with explicit claim boundaries, simulation evidence, WOD validation-CV auxiliary results, and packaged audit artifacts.

For CoRL:

Internal imitation-learning success can hide transfer-axis disagreement. We reproduce the mechanism under controlled proxy corruption and external AlpaSim transfer.

Do not pitch this as “the hybrid wins.”

The data does not support that.

Final Takeaway

The architecture work matters because it lets us localize failure:

- Spotlight shows geometry-grounded token selection can work
- DAgger learns the internal simulator boundary
- AlpaSim exposes proxy-state decision mismatch
- proxy perturbations reproduce the same failure internally
- WOD grounding shows frozen priors help only with the right head

The best claim is architectural and diagnostic:

robust AV evaluation needs grounded, per-axis transfer diagnostics,
not only offline accuracy or aggregate closed-loop pass rate.